

**UNIVERSITY OF SPLIT
FACULTY OF ELECTRICAL ENGINEERING,
MECHANICAL ENGINEERING AND NAVAL
ARCHITECTURE**

PROJECT

Pro-Build Construction

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MECHANICAL ENGINEERING AND NAVAL
ARCHITECTURE



Project from” Human-Computer Interaction”

This project was done by Ivan Zovko and Petar Vlaić for their subject of” Human-Computer Interaction”. Guided by professor Čagalj they made their web app using the knowledge they gained on the lectures.

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1 INTRODUCTION

Technology is on the constant rise, but its growth has been exponential in last few decades. That growth is not lacking in home-building technologies either. More technologies are introduced in houses and in the process of building or renovating them. That, combined with the growing population, has caused that everywhere you look there is a deficit of construction workers and crew, especially good ones. Motivated by the frustration we saw in our, but also our family and friends everyday lives, we have decided to find a solution to this problem with Pro-Build Construction.

Pro-Build Construction is a place where constant pressure of organizing multiple construction crew schedules, fearing if the job will be done well and fear of getting scammed is gone. Offering the customer to easily get an offer from one of the construction companies in a few clicks, allowing intuitive and easy project tracking, even online, and transparent company information with genuine ratings and comments as well as easy and fast complaints, Pro-Build Construction is for everyone hoping to build or rebuild a house, hotel or any kind of construction project of their choosing.

2 IDEA PITCH

In our recent past, both me and Zovko experienced firsthand what it's like to try and navigate in the mess of construction business, especially for a first timer. We often shared our frustrations during lecture breaks and then when we already agreed to be partners in the upcoming project, professor Čagalj pulled out a construction tool he just bought stating that he needs one and wanted to see how we would deal with a new product without knowing anything about it. This made us think if the problem is maybe more widespread than we originally thought. And indeed, a little bit of digging and questioning and we realized not only that a lot of people experienced this problem, but the hate for the difficulties on construction is so passionate that even those with no firsthand experience in it have developed a large fear of it. That was our "click" moment and from that point on we have vowed to solve this problem.

3 USER PERSONAS

The most widespread user persona we could think of was our everyday citizen, trying to build or renovate a home within a reasonable budget. But we didn't want to limit ourselves and decided to strive to account for everyone, even the 0.1%. With that in mind we thought of a few of them.

Persona 1: Marko “Entrepreneur Looking for New Investments”

Name: Marko

Age: 38

Location: Zagreb, Croatia

Occupation: Entrepreneur

Net Worth: ~4,000,000 EUR



Figure 3-1: User persona Marko

Background: Marko is a successful entrepreneur from Zagreb who grew up near the area where he plans to build his next project. His goal is not just to build a home, but to invest in a high-profile, premium property that can later be rented, sold, or used as a showcase for future investments. He wants a modern, digital-first approach without the outdated, manual hassle typical of construction in Croatia.

Goals:

- Build a high-profile, modern home as an investment
- Keep the entire process centralized and streamlined
- Work with professional, premium service providers
- Save time by automating as much of the process as possible

Pain Points

- Everything is done manually and offline
- Needs to search for every service separately (designers, engineers, permits, builders)
- Lack of digital tools for planning and communication
- Construction companies that are slow to respond or unprofessional

Needs:

- Fast, modern platform with a responsive and intuitive interface
- A centralized place that connects him to all the companies and services needed
- Automated communication, easy comparisons, and fast results
- Reliable and professional contractors

Persona 2: Anna “Retired High Net Worth Foreigner”

Name: Anna

Age: 60

Location: Vienna, Austria

Occupation: Retired (Husband is a well-known banker)

Net Worth: 20M+ EUR



*Figure 3-2: User
persona Anna*

Background: Anna and her husband fell in love with Dalmatian Zagora during a vacation and decided they wanted to retire there. They want a luxurious but traditional stone house with modern comforts, rooted in Mediterranean culture. Money is not an issue — time, trust, and quality are. They are worried about Croatian bureaucracy and do not want to rely on unknown local workers without guarantees.

Goals:

- Build a traditional stone home with modern features
- Move to a quiet village and enjoy Mediterranean lifestyle
- Fully outsource the process with minimal involvement
- Complete the project quickly and reliably

Pain Points:

- Slow and complicated Croatian bureaucracy
- Difficult to oversee construction from Austria
- Bad or unprofessional service from local companies
- Fear of delays, mistakes, or lack of transparency

Needs:

- Ability to track construction progress remotely

- A reliable, verified construction partner guaranteed by the platform
- Full project management without searching for contractors
- Fast construction with high-quality results, no matter the cost

Persona 3: Stipe “Local Family Man Building His New Home”

Name: Stipe

Age: 34

Location: Zagvozd, Croatia

Occupation: Local Clerk

Net Worth: ~200,000 EUR



*Figure 3-3; User
persona Stipe*

Background: Stipe recently got married and wants to build a modest but solid home. He has prepared the paperwork and a construction plan and is willing to invest time into coordinating the project — but he is afraid of scams, delays, and unreliable workers. His main priority is safety and quality, even if the project takes longer.

Goals:

- Build a safe, durable family home
- Work with honest and reliable builders
- Finish construction without losing money or nerves
- Stay within his financial limits

Pain Points:

- Fear of scammers and dishonest construction crews
- Workers who stop showing up or do minimal work
- Low-quality craftsmanship
- Hard to find trustworthy recommendations

Needs:

- A verified and trusted construction crew
- Guarantees that deadlines will be respected
- Protection from fraud and extra hidden costs

- Clear communication and progress updates

4 FIRST STAGE OF CODING

4.1 First idea of the sites look/Figma stage

We have thought of our layout a lot. At first our site was supposed to be all white and yellow. That way we would follow the mental model of “yellow=construction” and white-yellow combination would subconsciously associate the user with the “Njuškalo” site as a trusted and popular site among the people. That can be in our Figma visualization of low/high fidelity prototype that we presented back in December. The professor approved but added a comment that we could experiment with another color as the white and yellow could be of low contrast causing hard reading and unnecessary difficulty for the consumer.

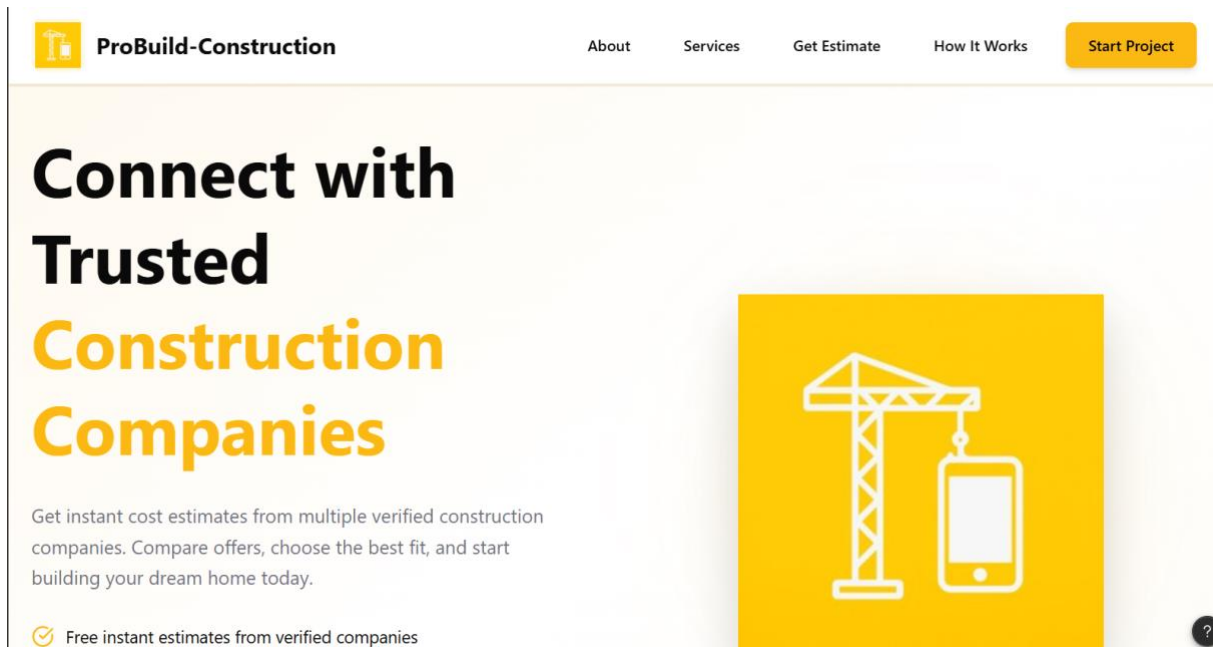


Figure 4-1: First visualization with Figma

4.2 New look and ideas

We opted for white on blue colors, with the primary color being blue with occasional yellow elements. That way we kept the yellow color, as it was the color of our sites logo, yet we fixed the low visibility issue of the white and yellow combination and kept the “construction kind of vibe” to the site as blue, white and yellow are often seen in construction sites.

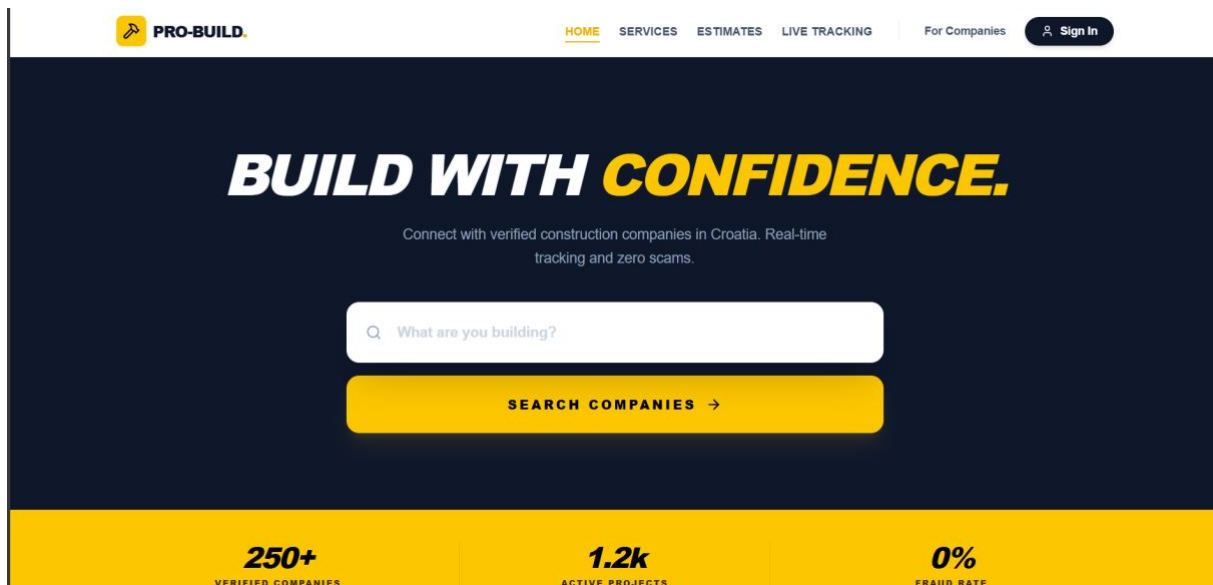


Figure 4-2 Websites new look

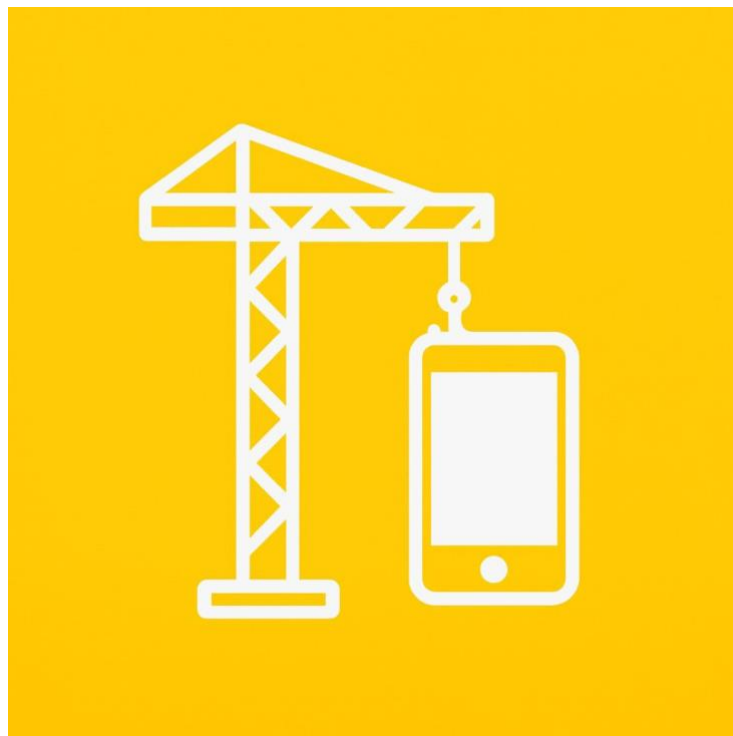


Figure 4-3 Pro-Build Construction logo

5 FINAL LOOK

We decided that the look could use a little “make up”. We concluded that the search bar would often make the user spend too much time on it as the user wouldn’t know what to search for. That’s why we opted for a different approach.

5.1 HOME

The home page includes 4 main quick routes to the services filtered to the users needs.

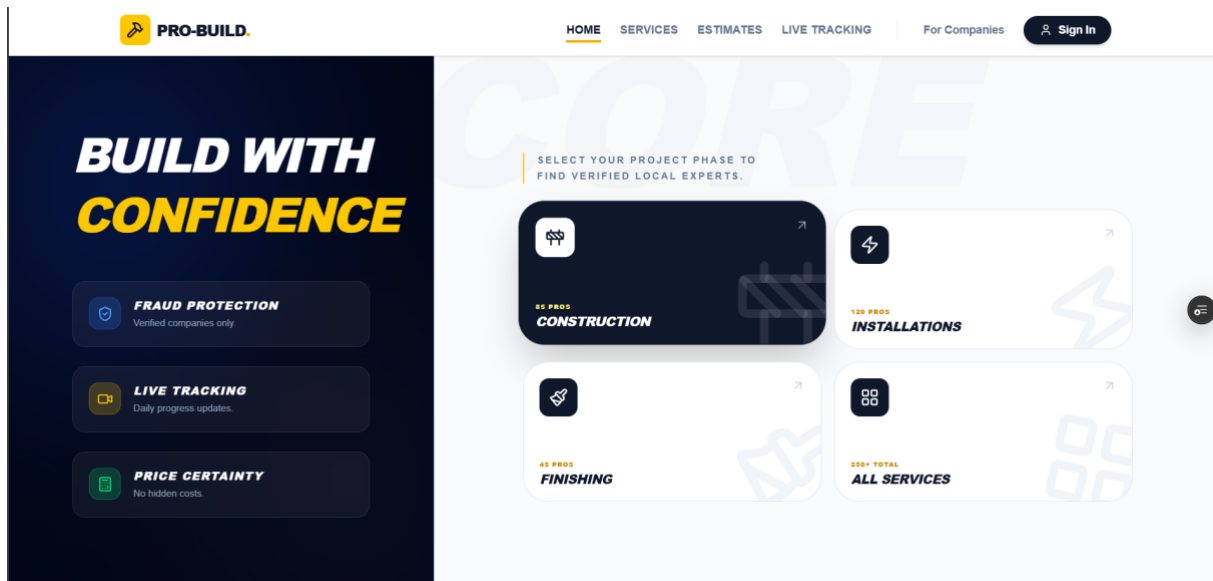


Figure 5-1 Final home page

At first our navigation bar was good, the active part of the navigation bar changed colors. However, listening to our professors advice we also added underlining to the currently active site.



Figure 5-2 Updated navigation bar

5.2 SERVICES

The SERVICES menu displays current available construction companies. If the companies aren’t listed the website will display the skeleton of the website, while the real data is loaded.

The customer can also filter the companies by location where their services can be offered. The companies also have easy to understand tags and locations under them. The blue mark is for verified companies that need to be verified by the website admin. Unfortunately, due to aesthetical reasons we chose to opt out of the Fitts’s “unlimited” button in the corner. Following Hick’s law, the distance between the filter and clickable companies is minimal. At first, we had plenty of filters and buttons but soon found out that they are rather confusing to our customers, especially the ones not so well educated in terms of IT technology. That’s why filters are preset by the website, and the client can choose which of those they want to choose. We enabled the client to set the location of their future project, filter by rating or alphabet and search only one specific need.

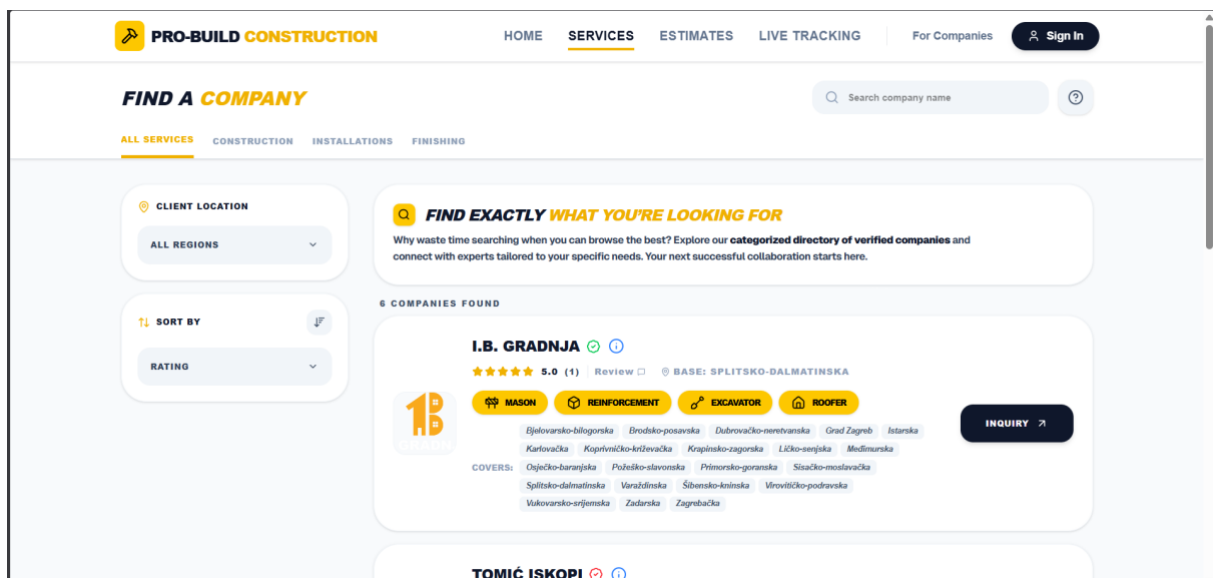


Figure 5-3 SERVICES page

5.3 ESTIMATES

This is what we initially intended to be our most used page, but it seems that, although a cool feature, plenty of people will use the SERVICES aswell. The idea behind the ESTIMATES is to turn the tides of looking for a construction company. Instead of sending dozens of emails to various companies and losing track of all the responses, you put your desired project in a few words, and the offer is sent out to the companies. Then they can follow you up and give you an offer and the user themselves decide on which one they will choose. On lectures our professor mentioned the need of suspense while being redirected. That’s why, even though the user needs to sign in first, we enable him “a taste” of the easy-to-use project calculator and then when they click “GET QUOTES” they are redirected to the sign in form.

The screenshot shows the 'ESTIMATES' page of the Pro-Build Construction website. The header includes the company logo, navigation links (HOME, SERVICES, ESTIMATES, LIVE TRACKING), and a 'Sign In' button. The main heading is 'PROJECT COST ESTIMATOR' with a subtitle 'CURRENT RATES FOR MATERIALS AND LABOR'. On the left, there is an 'FAQ' section with questions about estimate accuracy, VAT, and quality levels. The main content area is divided into three steps: 01 PROJECT TYPE (with buttons for HOUSE, APARTMENT, and RENOV.), 02 AREA SIZE (with a slider set to 100 M²), and 03 BUILD QUALITY (with buttons for BUDGET, STANDARD, and LUXURY). On the right, a 'LIVE ESTIMATE' summary shows a total investment of 165,000 €, an allocation chart for materials (62%), labor force (38%), and permits (10%), and a 'GET OFFERS' button.

Figure 5-4 Estimates page

If the user is signed in they can further describe their project and send a request.

The screenshot shows a 'PROJECT DETAILS' modal form. It has a title bar with a plus icon and a close button. The form contains three text input fields: the first is labeled 'T' and contains 'ŠIMUNIĆ'S LUXURY RECOVERY APARTMENT'; the second is labeled with a location pin icon and contains 'SPLIT, CROATIA'; the third is labeled with a list icon and contains 'HELP! I INJURED SULEJMANI, HE NEEDS TO RECOVER SOON OR HE WON'T BE ABLE TO COMPETE FOR BENFICA!!!'. At the bottom of the form is a dark blue button labeled 'REVIEW SUMMARY'.

Figure 5-5 Example of an offer

5.4 LIVE TRACKING

This feature enables signed in users to track and monitor their projects. They can either check their pending project estimates, check their past projects or look at the progress of their active projects with a progress bar.

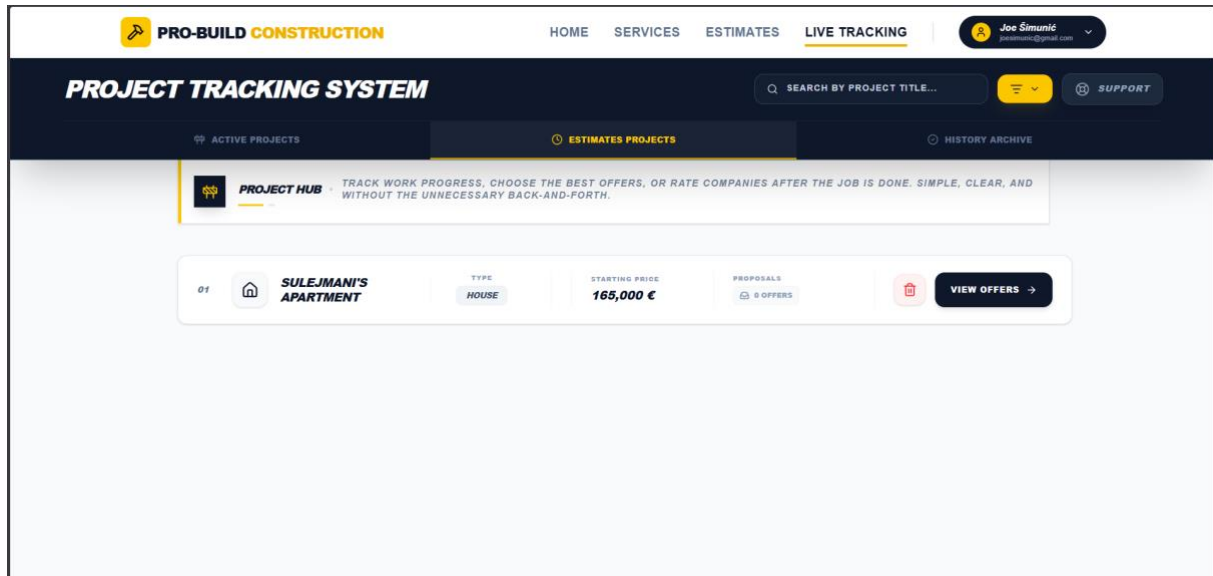


Figure 5-6 LIVE TRACKING page

Once logged in, the username is located on the top right model to adhere to the mental model of most of the websites that use log-in features. And if you click on it (the full name is clickable, not just the arrow to follow Fitts's principle) you get a drop down Sign Out option.

5.5 Sign in

The sign in form enables the user to sign in with either an existing Pro-Build Construction account or a google account. If they don't have an account, they can join in as a user.

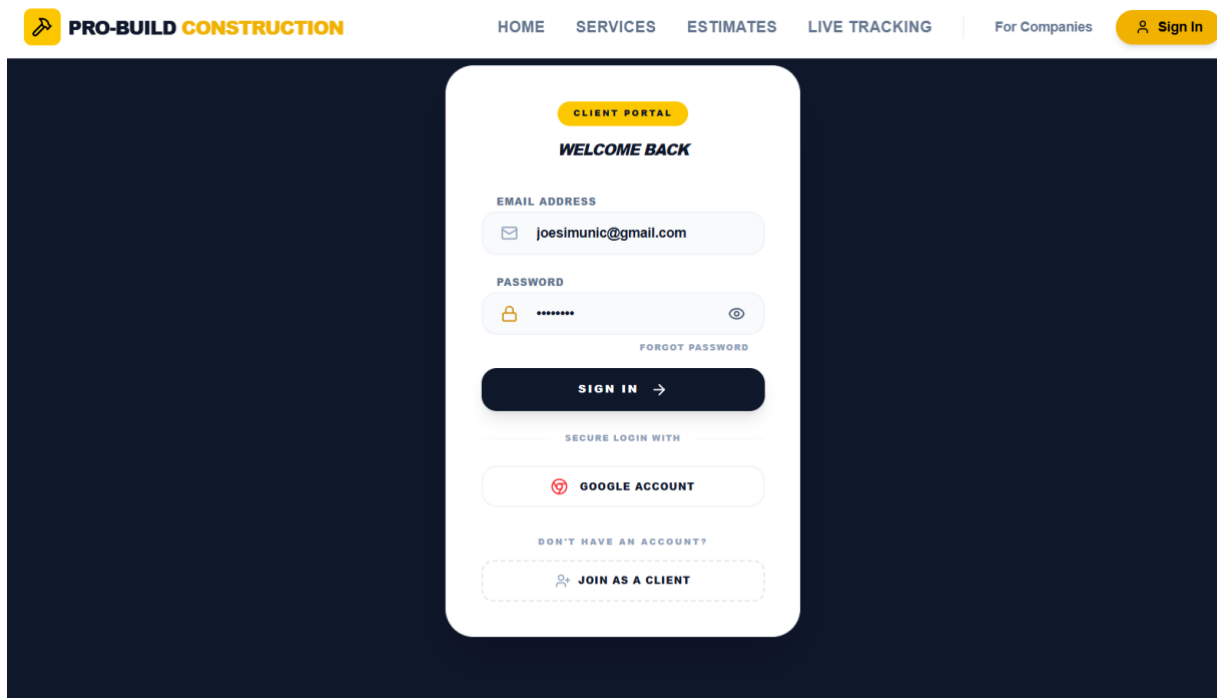


Figure 5-7 An example of log in details

The page itself has constraints, as to prevent the user from entering wrong input. Password character limit is set to 6 minimum, and the email must have a “@” character in it.

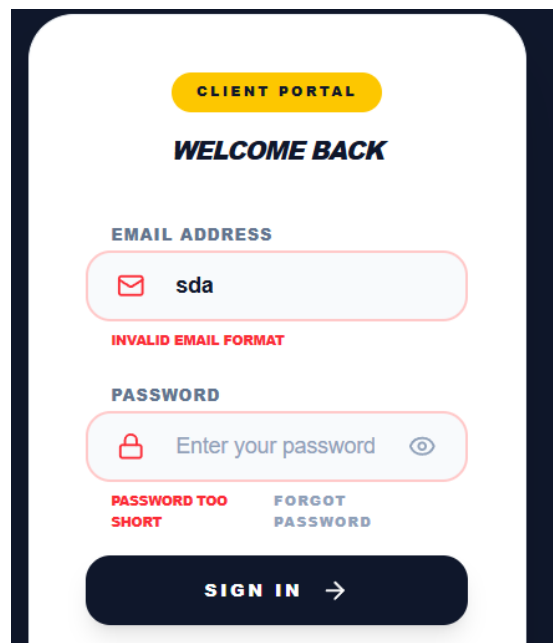


Figure 5-8 Constraints principles

5.6 For Companies

Companies have a separate log in/register form. Theirs register form is much longer than that of a regular user. They need to set up all their details and at the end have to upload documents to verify their identity.

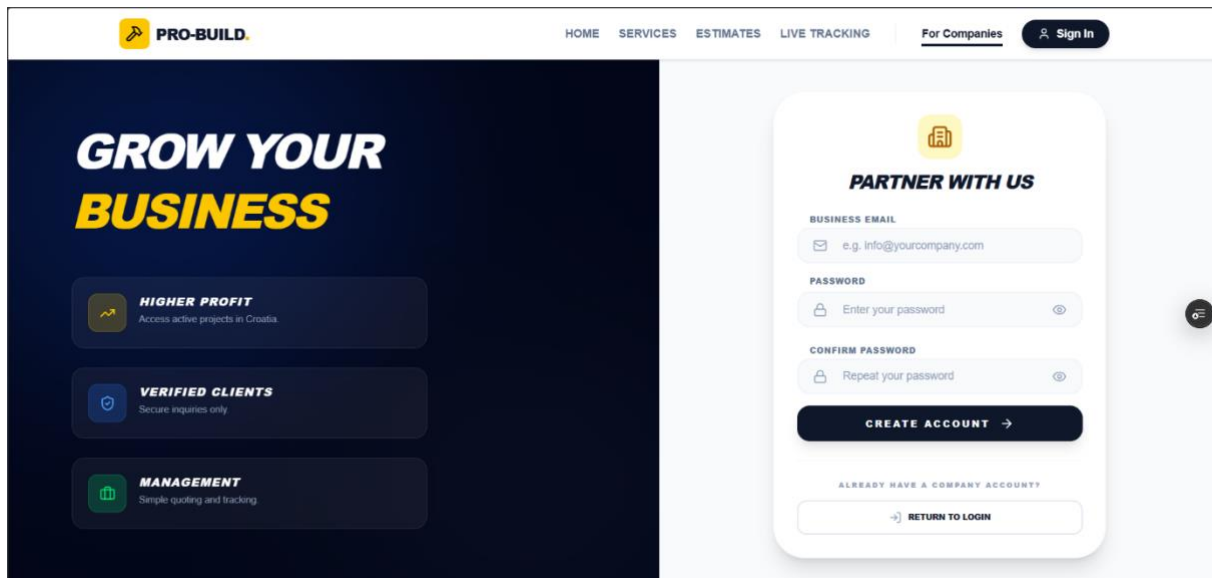
The screenshot displays the PRO-BUILD website interface. The top navigation bar includes links for HOME, SERVICES, ESTIMATES, LIVE TRACKING, and a 'For Companies' section with a 'Sign In' button. The main content area is split into two panels. The left panel, with a dark blue background, features the heading 'GROW YOUR BUSINESS' and three benefit cards: 'HIGHER PROFIT' (Access active projects in Croatia), 'VERIFIED CLIENTS' (Secure inquiries only), and 'MANAGEMENT' (Simple quoting and tracking). The right panel, with a light gray background, contains the 'PARTNER WITH US' registration form. This form includes input fields for 'BUSINESS EMAIL' (with a placeholder 'e.g. info@yourcompany.com'), 'PASSWORD', and 'CONFIRM PASSWORD'. Each password field has a lock icon and a toggle for visibility. Below the form is a 'CREATE ACCOUNT' button with a right arrow, and a 'RETURN TO LOGIN' link for existing users.

Figure 5-9 For Companies login

When properly registered the company can login into their account. Once logged in they have a separate page where they can overview their current offers and projects they picked up, read and answer inquiries, view available projects, be in direct contact with their customers via messages, track their projects and check and review their company's profile.

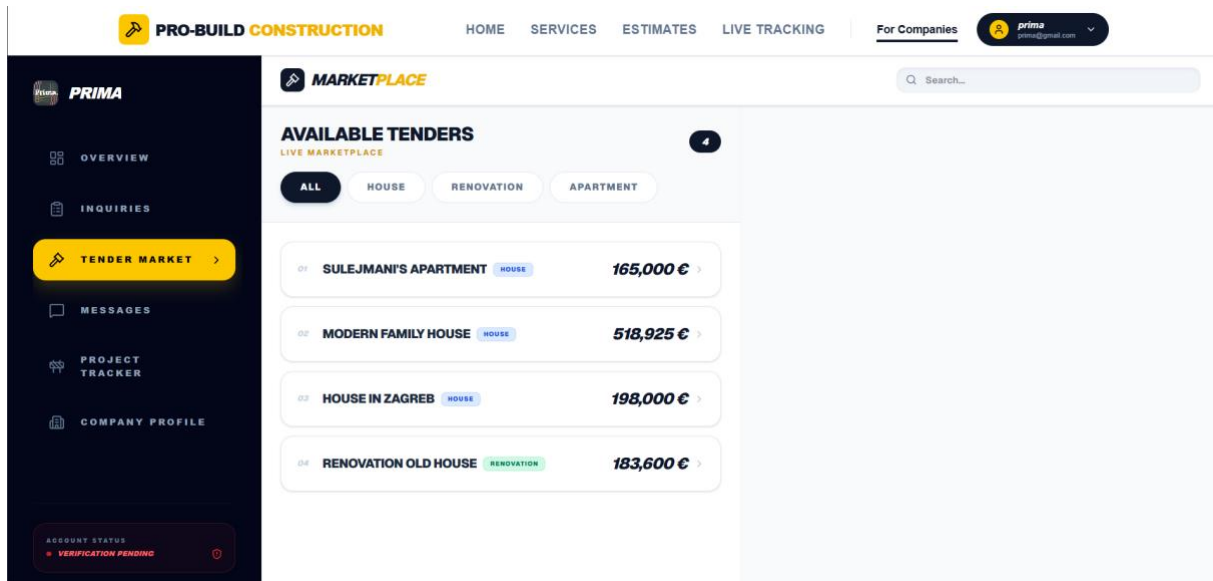


Figure 5-10 For Companies page

5.6.1 Constraints of the For Companies page

The company is required to fill all the fields. The PIN number must be 11 digits exactly. Password can't be shorter than 6 symbols. At the end they are required to upload documents in necessary formats (pdf, png or jpg).

BUSINESS DETAILS

Official company information.

COMPANY NAME

PIN

BUSINESS ADDRESS

ADDRESS IS REQUIRED

HQ COUNTY

Figure 5-11 For Companies constraints

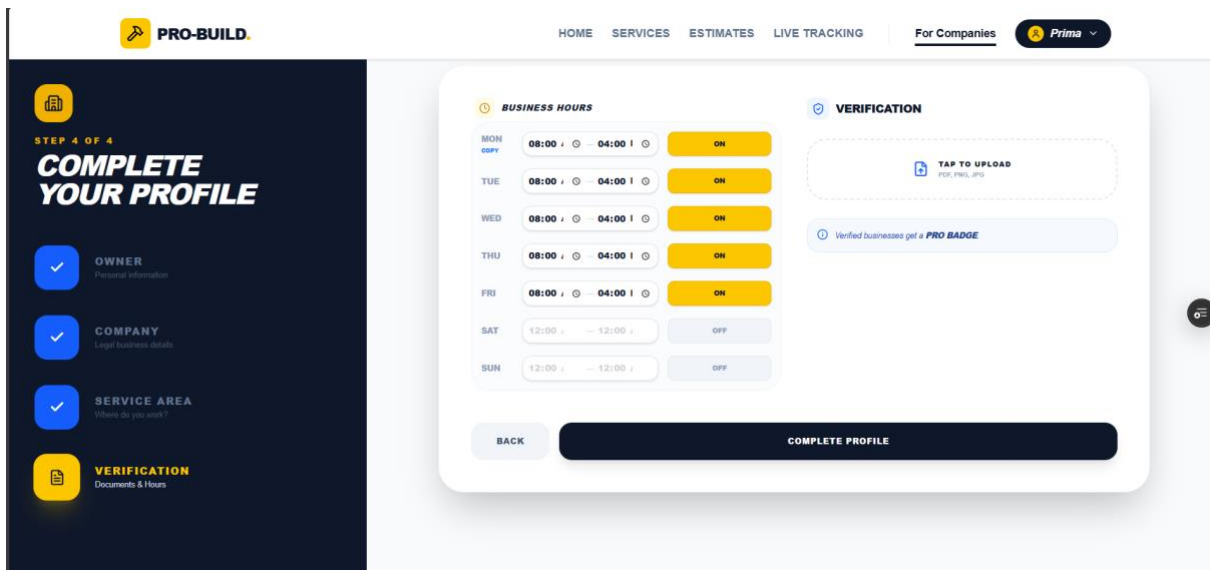


Figure 5-12 Verification files constraint (in the file to upload section)

5.7 Small screen optimization

The website has been adjusted for smaller screens such as smartphones. The usual navigation bar is replaced with hamburger bar and the content of the website is adjusted for vertical screens.

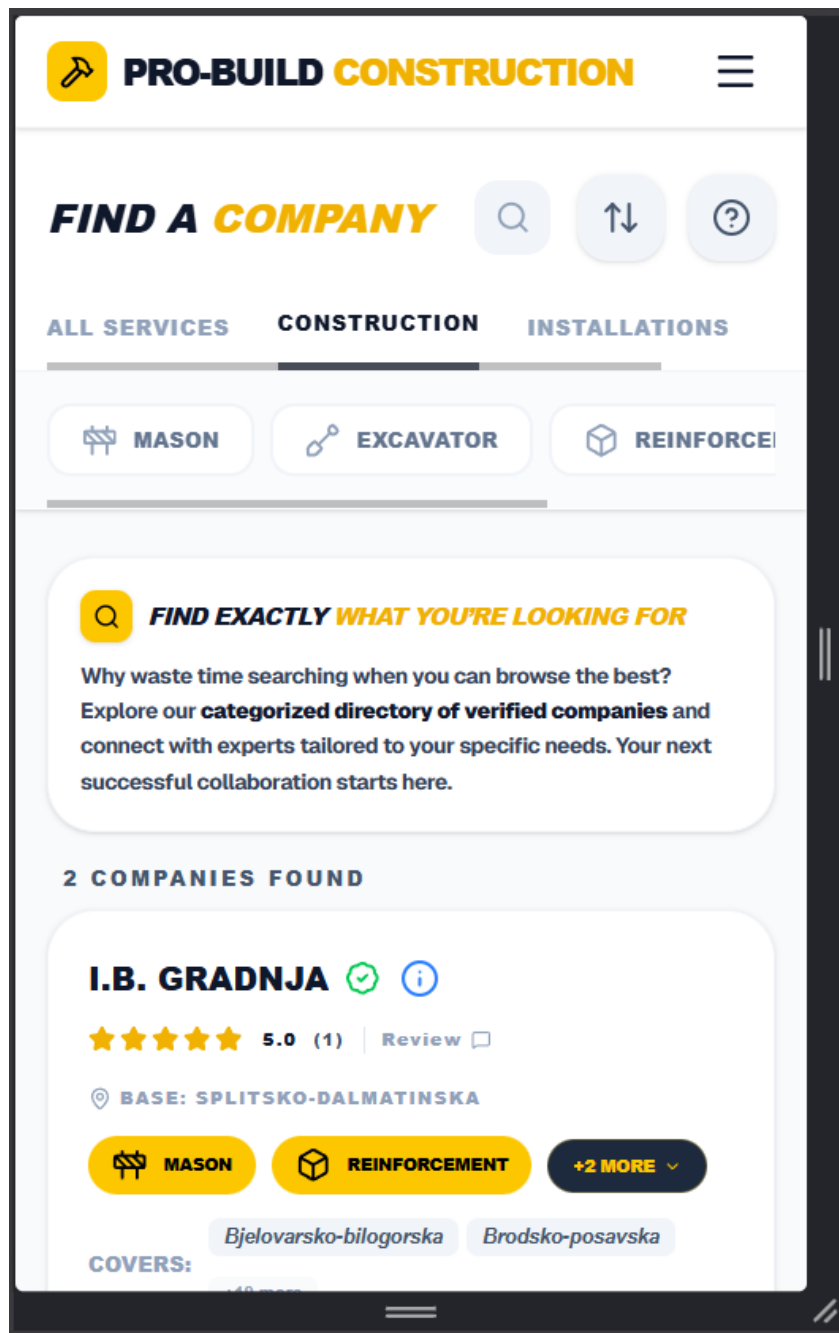


Figure 5-13 Small screen optimization

6 HCI RULES THAT WERE RESPECTED

6.1 Fitts' law

All buttons were made big enough to click easily. We didn't implement "infinity buttons" in the corners due to design aesthetic.

6.2 Hook's law

The number of actions the user can perform is limited to only the necessary functions. Those functions are easily understandable and easy to spot due to Fitts's law.

6.3 Nielsen's 10 Usability Heuristics

- We used common mental models of other websites that use similar layouts (such as njuskalo.hr for services page or facebook.com for log in)
- The colors, theme and fonts are consistent throughout the whole website across the pages.
- The site itself is quick but in case the website fails to load instantly we added suspense that lets the user know the action was registered and gives him a loading screen until the data loads.
- The user is warned about deleting important information and data beforehand.
- The user can easily organize and filter their projects or companies in the SERVICES and LIVE TRACKING pages.
- Shortcuts for expert users are provided on the HOME page (they jump to filtered companies in the SERVICES page)
- Design is set to minimum to avoid overloading the user with unnecessary information

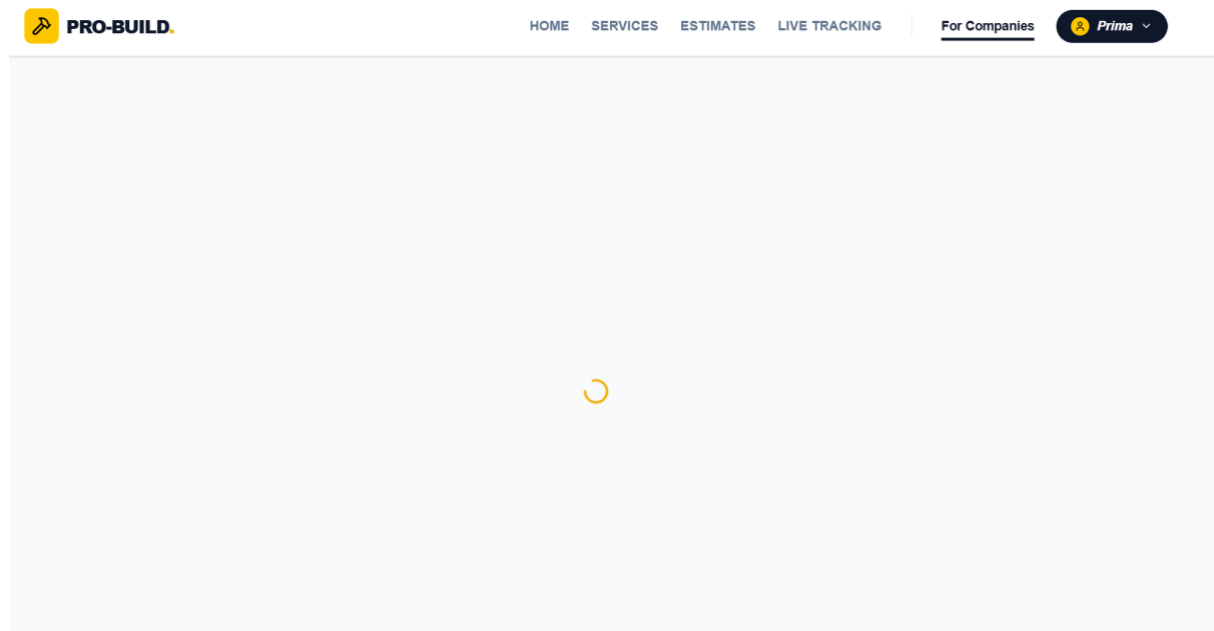


Figure 6-1 Example of “faking” the instant response

6.4 Constraints

They are put throughout the page, mostly on the log in parts of the website and have been explained in the previous chapter (such as chapter 5.6.1)

6.5 Use of colors and fonts

Colors of white and yellow were the first choice as they follow construction company themed colors and associate the user to the njuskalo.hr website. However, due to professors input we opted to pick blue for the background to make the website easier to read. Also blue, yellow and white avoid the most common forms of color blindness, to the colors of red and green (8% of male population and 0.5% of female population have it). Fonts are consistent throughout the whole website to follow the rule of consistency.

6.6 Added highlighted search

We noticed intuitively, when users with higher computer skills use the website, that when they search for items, they expect the typed in words to be highlighted so they know their input is correct. That has been added to the website for a smoother experience.

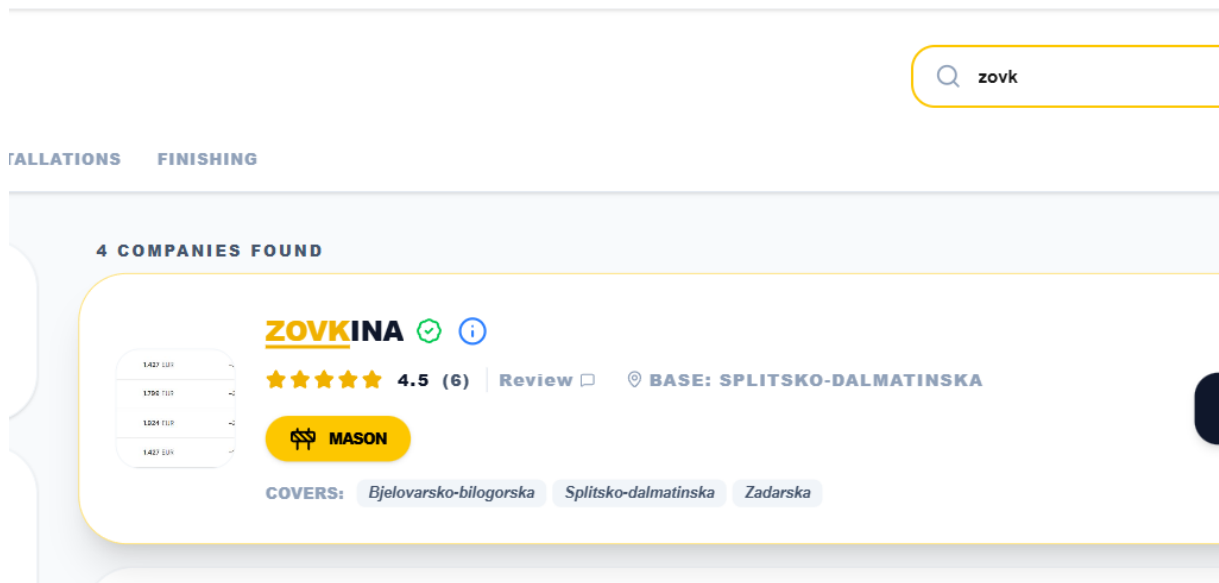


Figure 6-2 Highlighted search

6.7 Hover indicating the activity

We also follow the advice of our professor which he pointed out during the lecture that when a button or a function is hovered, it would be good to have a small window appear and say what it does.



Figure 6-3 Hovering the function

6.8 Optimization for smaller screens

The website has been optimized for smaller screens. The navigation bar is replaced with a hamburger and the website data has been adjusted accordingly as to not overload the user.

6.9 Warning before big actions

When user is performing a big or irreversible action, the website warns him about it and makes sure that there are no miss clicks.

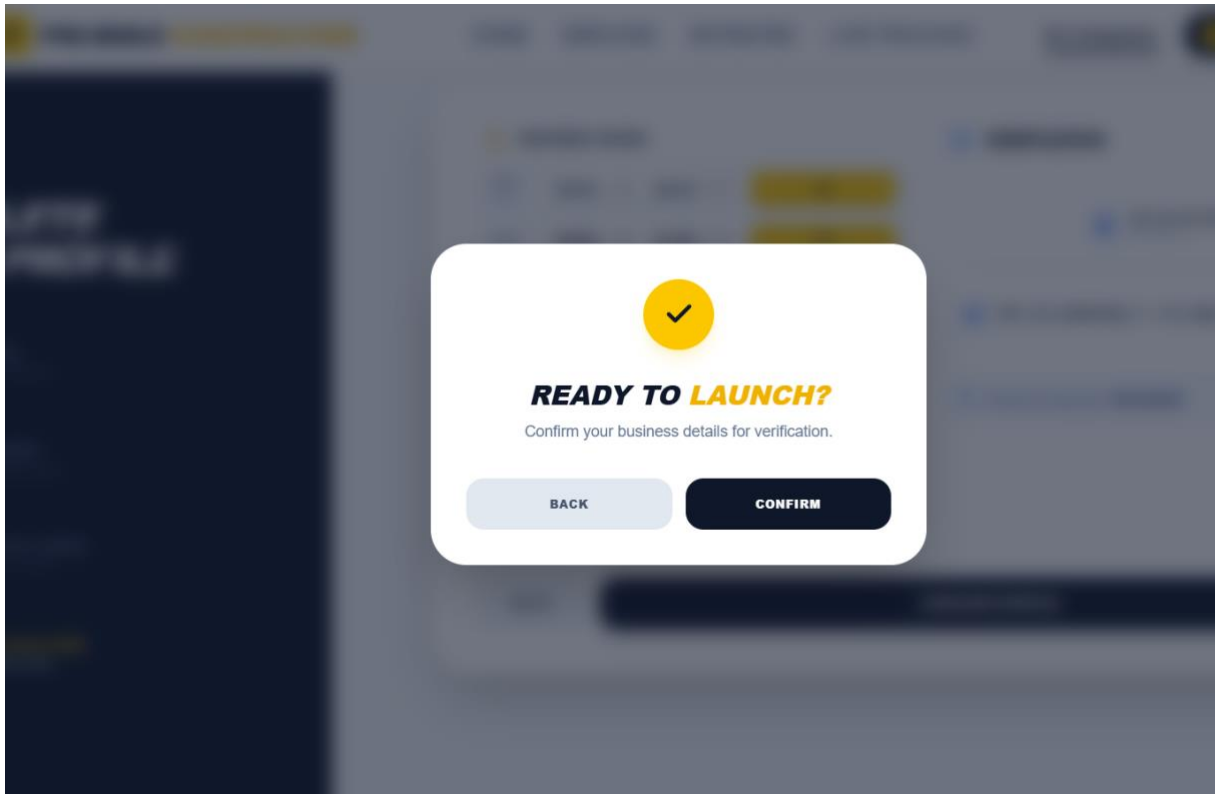


Figure 6-4 Warning before big action

6.10 Added pagination

To avoid the user being overstimulated by bunch of data at once and to reduce the data needing to be grabbed from the server at once we added pagination. It is both easier for the user to navigate such website and for the website itself to function.

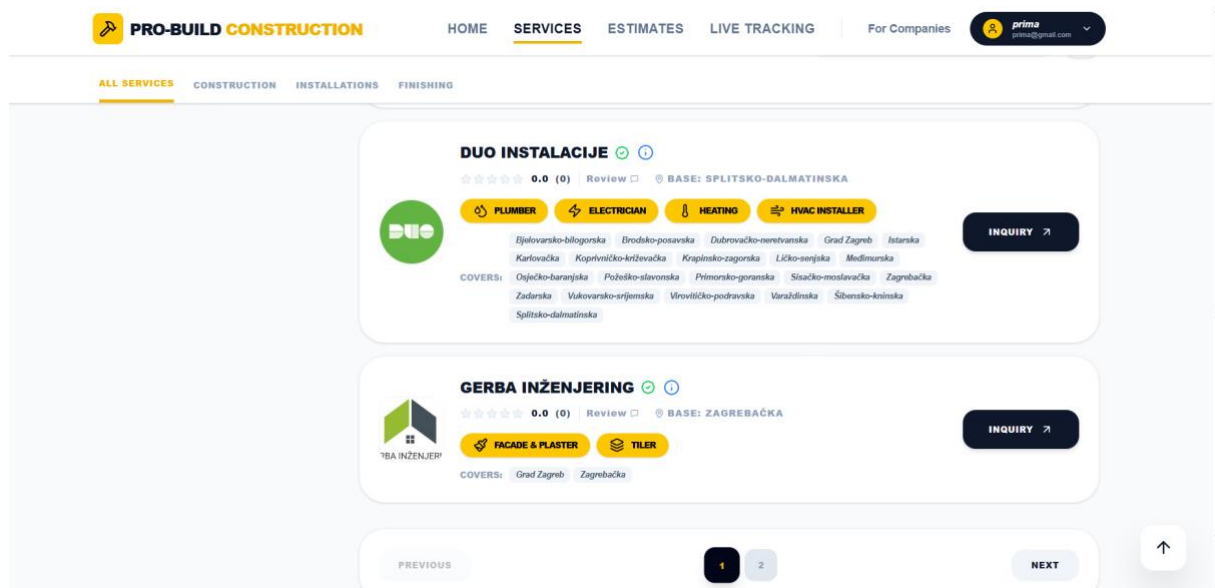


Figure 6-5 Pagination

6.11 Visible changes

Visible changes have been added to the website so the user knows when their change comes through. This has been added to avoid user frustration of being in suspense.

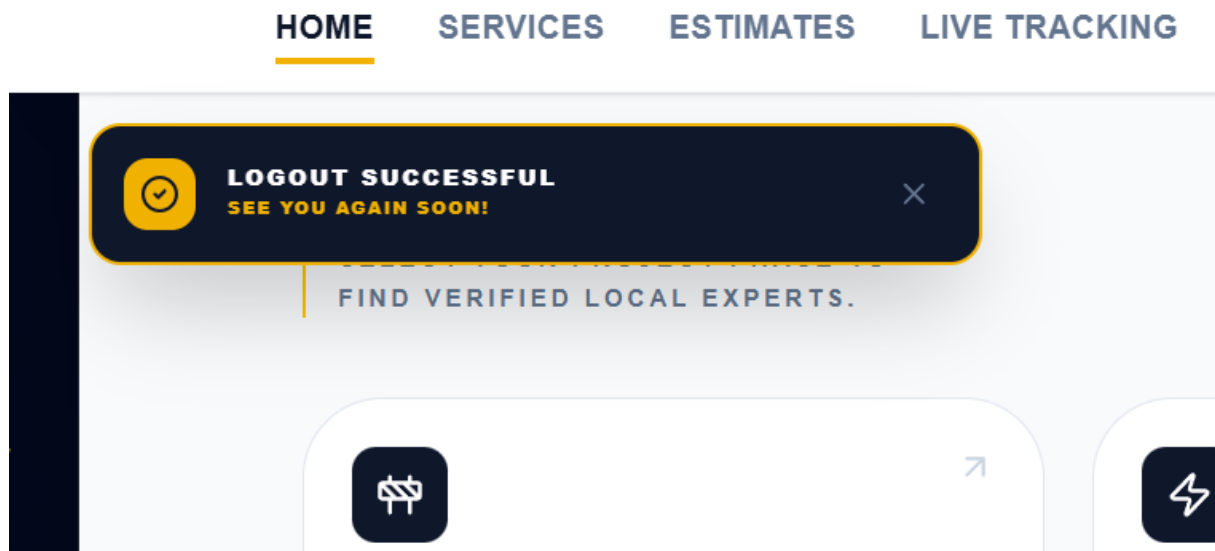


Figure 6-6 Visible changes of logging out

6.12 CMS

CMS has been added to the website for easier data changes, website updates and managing the website overall. We added admin login information that is typed in the regular login form. In the admin page the user (in this case admin) can manage content with ease.

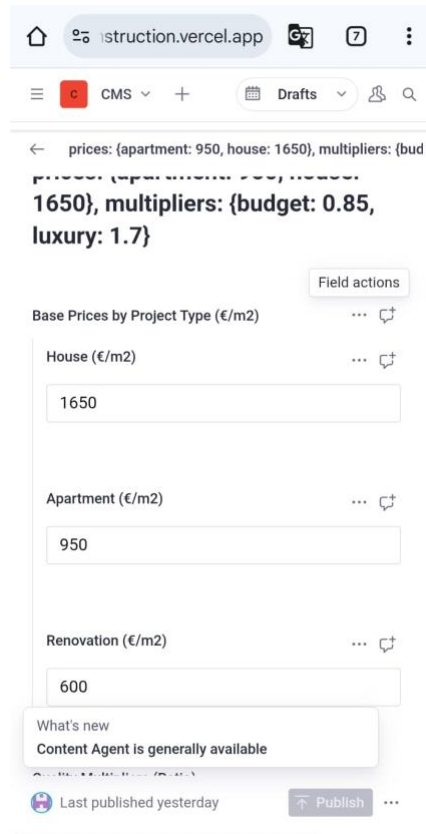


Figure 6-7 CMS 1

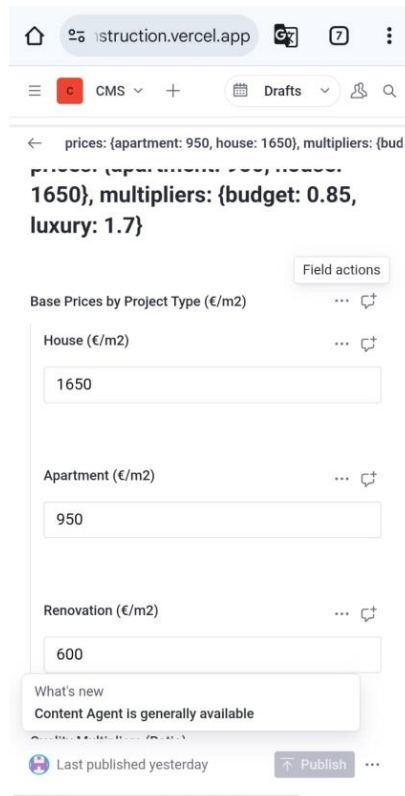


Figure 6-8 CMS 2

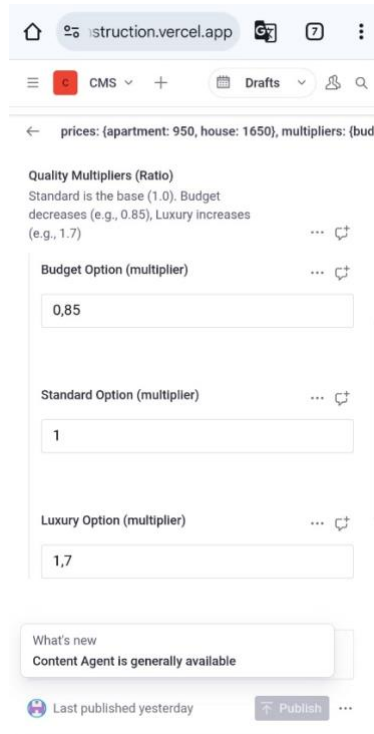


Figure 6-9 CMS 3

7 PageSpeed analysis

7.1 Desktop analysis

PageSpeed analysis provided us with good feedback on the desktop version of the website, with a good response time. Only critic was bad color contrast, which might be a case but due to the short schedule is left unattended. In websites further updates it might be changed.

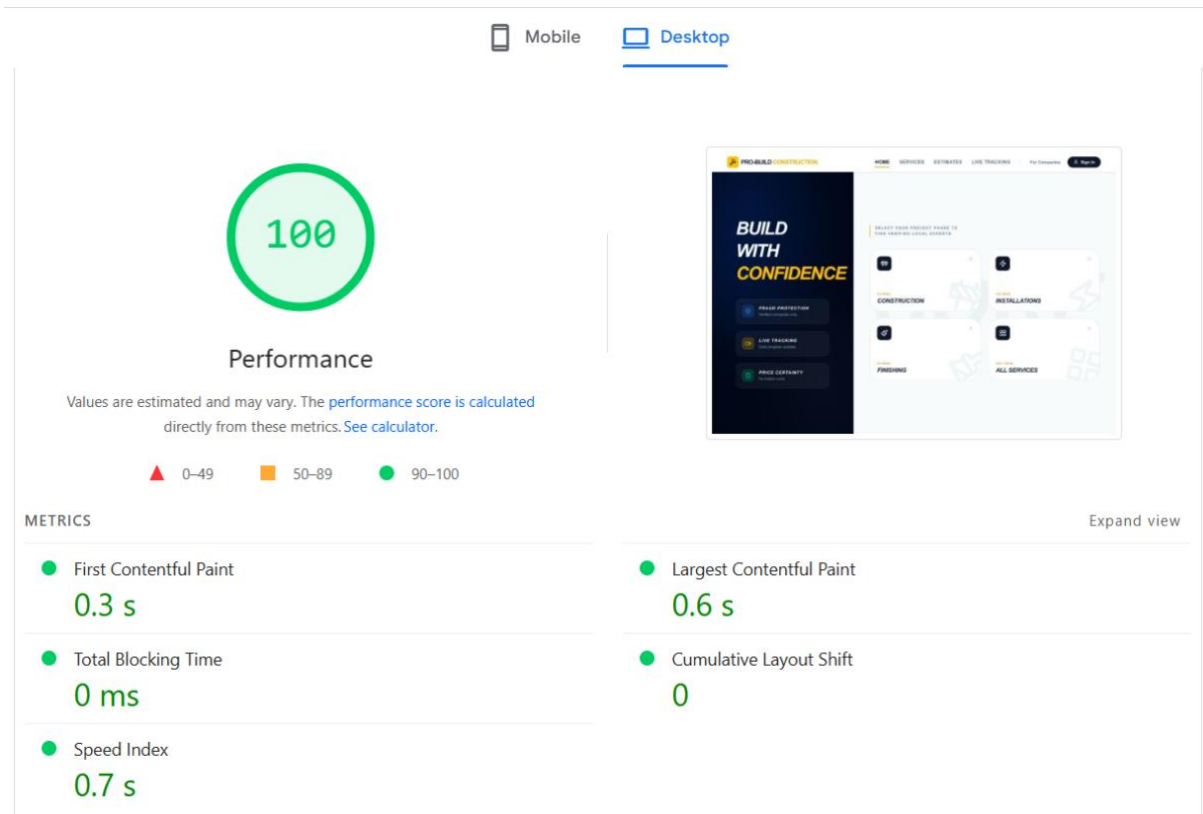


Figure 7-1 Desktop analysis

7.2 Mobile analysis

The mobile version was a bit slower and had the same color issues as the desktop version, so it got lower scores than the desktop version.

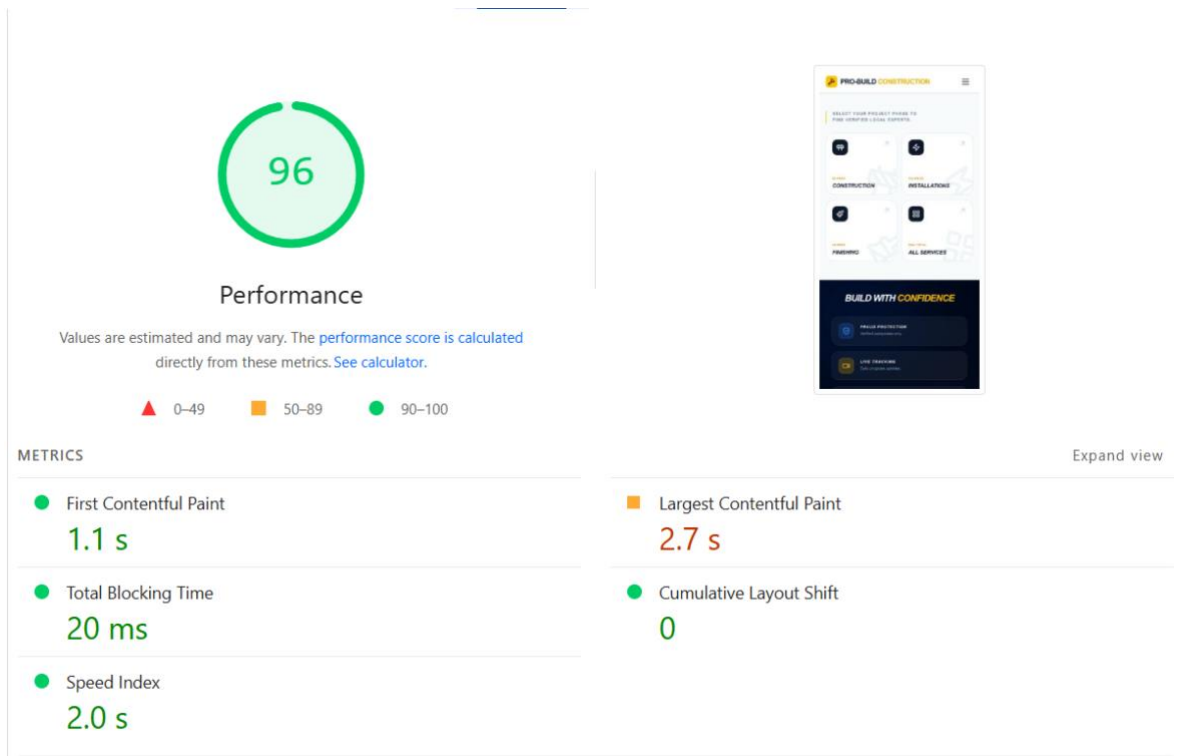


Figure 7-2 Mobile analysis

LITERATURE

Various LLM languages, mostly Gemini and professors lectures.

- <https://gemini.google.com/>
- <https://chatgpt.com>

CONTRIBUTIONS

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List of abbreviations

HCI	Human-Computer Interaction
LLM	large language model

ABSTRACT AND KEYWORDS

Abstract

The website that was created and described in this paper was made to resolve the constant and never-ending struggles of navigating through the construction work hell. After creating a wide pool of user personas, the task was to make the website intuitive and easy to use for all of them. With the graphically modern design of the home page the users would easily land on their desired places, without having to “learn” the website or plan for it. The website lands them on their desired preferences and filters construction companies just for them. If a user needs to make a project from scratch and is overwhelmed by the mountain of work that needs to be done, they can simply go to the estimates page and there they can get a visualization and approximation of the whole project served to them by licensed professionals. For some of the functions log in was used. The website recognizes two different types of accounts: one for companies and one for users. Users can easily overwatch and monitor their current projects, check up on their pending project requests and look over their past projects.

Keywords

HCI, construction, home building, renovation, investment